ModaNet: Synthesizing Missing MRI Sequences with GANs Using Patch-Based Discrimination

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***Abstract*—Magnetic Resonance Imaging (MRI) provides crucial diagnostic information for brain tumor assessment, but complete multimodal sequences are often unavailable due to clinical constraints, cost limitations, and patient-related factors. This study presents ModaNet, a conditional Generative Adversarial Network (cGAN) approach for synthesizing missing MRI sequences in brain imaging. The proposed method employs a U-Net generator architecture combined with a patch-based discriminator to generate high-quality, realistic MRI sequences from available modalities. The proposed framework was evaluated on the BraTS2024 dataset, demonstrating superior performance compared to existing methods. For T2-weighted image synthesis, ModaNet achieved a Peak Signal-to-Noise Ratio (PSNR) of 37.39 and a Structural Similarity Index Measure (SSIM) of 0.978, significantly outperforming pix2pix (26.16 PSNR, 0.821 SSIM), MedGAN (25.62 PSNR, 0.805 SSIM), and pGAN (27.40 PSNR, 0.846 SSIM). For T2-FLAIR synthesis, ModaNet achieved an SSIM of 0.969, substantially exceeding the SSIM of transformer-based methods (0.859). The quantitative results demonstrate the effectiveness of our approach in generating missing MRI sequences, offering a valuable tool for enhancing diagnostic capabilities and supporting AI-driven brain tumor analysis in clinical settings.**

***Index Terms*—Brain MRI, GANs, Medical Image Synthesis, Missing Modalities, Glioblastoma**

I. Introduction

Magnetic Resonance Imaging (MRI) demonstrates remarkable versatility through its wide range of sequences and imaging parameters, each designed to underline specific tissue characteristics and pathological changes. This flexibility gives rise to different MRI modalities, such as T1-weighted, T2-weighted, diffusion-weighted imaging (DWI), and fluid-attenuated inversion recovery (FLAIR) [1], each providing complementary diagnostic insights. In clinical practice, acquiring multi-modal MRI datasets is standard, as the integration of these varied imaging perspectives allows for a more nuanced and comprehensive evaluation of underlying pathologies, eventually supporting more accurate diagnosis and informed therapeutic decision-making. The International Brain Tumor Segmentation (BraTS) [8] challenge has played a pivotal role in advancing algorithms for the segmentation of brain tumors using multiple MRI sequences. However, obtaining complete MRI sequences is often impractical due to factors such as cost, data corruption, and patient movement or allergies to contrast agents.

To address this issue, we propose a Generative Adversarial Networks (GANs) based approach to synthesize missing MRI sequences. The proposed method employs a custom U-Net architecture for the generator and a patch-based discriminator to produce high-quality, realistic MRI sequences. This approach enhances diagnostic capabilities and supports artificial intelligence-driven analysis, thereby facilitating more effective and efficient medical diagnostics. The proposed method is evaluated using the BraTS 2021 dataset [8], demonstrating promising results in generating missing MRI sequences. Quantitative evaluations, utilizing metrics such as Structural Similarity Index Measure (SSIM), and Peak Signal-to-Noise Ratio (PSNR) indicate the efficacy of our method. By synthesizing missing MRI modalities, our approach offers a valuable tool for improving the diagnosis and treatment planning of brain tumors, ultimately contributing to the advancement of medical imaging technologies.

The remainder of this paper is structured as follows: Section II reviews the relevant literature for this research. Section III outlines the dataset pre-processing technique employed, details the proposed methodology including the architectural components and model configurations, and presents the experimental setup, evaluation metrics, and results. Finally, Section IV concludes the paper and highlights potential directions for future work.

II. Related Work

Magnetic Resonance Imaging (MRI) plays a significant role in the diagnosis, treatment planning, and follow-up of patients with brain tumors due to its non-invasive and radiation-free nature. The International Brain Tumor Segmentation (BraTS) [8] challenge has significantly contributed to the development of numerous AI algorithms for accurately and efficiently segmenting glioblastoma sub-compartments using four structural MRI sequences: T1-weighted pre-contrast (T1), T1-weighted post-contrast (T1-Gd), T2-weighted (T2), and T2 fluid-attenuated inversion recovery (T2-FLAIR). However, acquiring all four necessary structural MRI sequences may not always be practical due to factors such as cost, data corruption, patient movement, or allergies to contrast dye.

Shen and Gao [2] investigated brain tumor segmentation on MRI with missing modalities, demonstrating the challenges and potential solutions for handling incomplete MRI data in clinical applications. Zadeh *et al.* [3] provided a comprehensive review of GANs for brain image synthesis, highlighting the various GAN architectures and their applications in medical imaging. Their survey covered different approaches for generating high-quality brain images and synthesizing missing modalities. Xie *et al.* conducted a comprehensive survey on cross-modality neuroimage synthesis, exploring various methods for generating missing neuroimaging modalities [4]. Their work provided insights into the current state-of-the-art approaches and identified key challenges in cross-modal synthesis.

Hamamci [5] implemented and utilized an open-source GAN approach that takes any three MRI sequences as input to generate the missing fourth structural sequence. This approach, integrated into the Generally Nuanced Deep Learning Framework (GaNDLF), demonstrated promising results in synthesizing high-quality and realistic MRI sequences. The study utilized the BraTS 2021 dataset [9], which comprises images of 1,250 glioblastoma patients, and employed a Pix2pixHD model adapted to take three inputs and output one, thereby leveraging all available information to generate the missing modality. The experimental approaches in these studies typically involved training multiple generators for synthesizing different missing MRI sequences. The performance of these approaches was quantitatively evaluated using metrics such as Learned Perceptual Image Patch Similarity (LPIPS), Normalized Mutual Information (NMI), Fréchet Inception Distance (FID), and SSIM. Results consistently indicated that the developed models were capable of producing high-quality and realistic images, closely resembling ground truth images.

Conte *et al.* present a compelling study on the use of GANs to synthesize missing T1 and FLAIR MRI sequences for brain tumor segmentation models [6]. This research is particularly relevant in scenarios where missing MRI sequences pose a significant obstacle to the application of deep learning models that require multiple MRI inputs. They highlight the potential of GANs in enhancing the applicability and robustness of deep learning models in medical imaging, particularly in contexts where complete MRI sequences are unavailable. The study also explores the generalizability of the models by testing them on independent datasets, demonstrating their potential for broader clinical applications.

In the study by Pan *et al.*, a novel approach is introduced that leverages transformer-based architectures for the synthesis of T2-weighted MRI images from T1-weighted images [7]. This work addresses the common clinical challenge of missing MRI modalities, which can impede comprehensive diagnostic evaluations. The authors propose a supervised deep learning method that utilizes a transformer-based encoder within a conditional Generative Adversarial Network (cGAN) framework. The model is designed to translate 2D MR images from T1 to T2 by incorporating adjacent slices to enrich spatial prior knowledge. The effectiveness of this approach is validated on both a private dataset and a public dataset, demonstrating superior performance over state-of-the-art methods in terms of quantitative metrics such as PSNR and SSIM. The study highlights the potential of transformer-based models in medical image synthesis, offering a promising solution to the modality-missing issue in clinical MRI practice. The integration of adjacent slice information and the use of a transformer-based encoder are shown to significantly enhance the quality and accuracy of synthesized images, thereby facilitating more reliable and comprehensive diagnostic processes.

Recent advances in this field have demonstrated the potential of GANs and Transformers to generate missing modalities in brain MRI, offering valuable tools for enhancing the diagnosis and treatment of patients with glioblastoma. The proposed ModaNet framework provides a solution capable of generating multiple MRI modalities from any combination of available inputs. This addresses a critical limitation in existing methods where clinical scenarios often involve multiple missing modalities simultaneously. Our patch-based discriminator design explicitly addresses computational limitations by enabling effective local feature discrimination while maintaining computational efficiency.

III. Materials and Methods

A. Dataset

We used BraTS 2024 dataset [10] which includes images from patients diagnosed with glioma, a prevalent and aggressive form of brain cancer. It is designed to facilitate the development and evaluation of automated segmentation models. The dataset comprises multiparametric MRI (mpMRI) scans, including pre-contrast T1, T1-Gd, T2, and FLAIR sequences. The data was contributed from multiple academic medical centers and includes detailed annotations of tumor sub-regions, such as enhancing tissue (ET), surrounding non-enhancing FLAIR hyperintensity (SNFH), non-enhancing tumor core (NETC), and resection cavity (RC). The BraTS 2024 dataset [10] is notable for its focus on post-treatment imaging, which presents unique challenges due to treatment-related changes such as resection cavities, blood products, and post-radiation inflammation. The inclusion of these complexities makes this dataset particularly valuable for developing and evaluating models that aim to improve the segmentation and analysis of post-treatment brain tumors.

B. Preprocessing

The preprocessing of the MRI data involved several critical steps to ensure the quality and consistency of the input data for the GAN model. Initially, the MRI scans were loaded using the nibabel library, which is a Python package for accessing a variety of neuroimaging data formats. Each MRI modality was resampled to a uniform size of 256×256 pixels to standardize the input dimensions. For each patient, the central slices of the MRI volumes were selected to focus on the most relevant regions of the brain. Specifically, a predefined number of slices per patient, centered around the middle slice, were extracted and preprocessed. Each slice underwent normalization to scale the pixel intensities to a range between 0 and 1, which aids in stabilizing the training process of the neural network. This normalization was followed by resizing the slices to the target dimensions using interpolation to maintain the structural integrity of the images.

C. Proposed Method

This study employs a Conditional GAN (cGAN) to synthesize specific MRI modalities from other given modalities. The methodology leverages the capabilities of cGANs to handle multi-modal data and generate high-quality images, which is particularly useful in medical imaging tasks.

The overall architecture of the proposed system is illustrated in Fig 1. The generator architecture employed in this study is based on a U-Net structure [11]. The U-Net architecture is characterized by its symmetric design, featuring a contracting path (encoder) followed by an expansive path (decoder). This design allows the network to capture context at different scales and precisely localize features, making it highly effective for generating detailed and realistic images.

The contracting path of the U-Net consists of several convolutional blocks, each designed to downsample the input image while increasing the number of feature channels. Each block in the contracting path typically includes a convolutional layer with a 4×4 kernel size and a stride of 2, which effectively halves the spatial dimensions of the input feature maps. This downsampling process is followed by batch normalization to stabilize and accelerate the training, and a LeakyReLU activation function to introduce non-linearity into the model. The use of LeakyReLU helps mitigate the vanishing gradient problem, thereby facilitating the training of deeper networks.

At the bottleneck of the U-Net, the feature maps are processed by a convolutional layer that further increases the depth of the feature representation. This layer serves as a bridge between the contracting and expansive paths, capturing the most abstract and compressed representation of the input image.

The expansive path, or decoder, mirrors the contracting path but in reverse. It consists of several upsampling blocks that use transposed convolutional layers to increase the spatial dimensions of the feature maps. Each block in the expansive path typically includes a transposed convolutional layer with a kernel size of 4×4 and a stride of 2, which doubles the spatial dimensions of the input feature maps. This upsampling process is followed by batch normalization and a ReLU activation function. The expansive path also incorporates skip connections from corresponding layers in the contracting path, which concatenate feature maps from the encoder to the decoder. These skip connections help the network retain fine-grained details and spatial information, which is crucial for generating high-quality images.

The final layer of the generator is a transposed convolutional layer that maps the high-dimensional feature representation to the desired output image. This layer is followed by a Sigmoid activation function to ensure that the output pixel values are within the valid range of [0, 1].

In general, the U-Net generator architecture is designed to effectively capture and reconstruct detailed features from input MRI modalities, producing realistic and high-quality output images. The combination of downsampling and upsampling paths, along with skip connections, enables the network to generate images that closely resemble the target modalities, making it a powerful tool for medical image synthesis.

1) Discriminator Architecture: The discriminator architecture in this cGAN is designed to distinguish between real and generated images, playing a crucial role in the adversarial training process. Unlike the generator, which focuses on creating realistic images, the discriminator acts as a critic, refining its ability to differentiate between authentic MRI images and those synthesized by the generator.

The architecture begins with an initial convolutional layer that takes the concatenated input, which includes both the input modalities and the generated or real target images. This layer employs a 4×4 kernel size and a stride of 2, effectively reducing the spatial dimensions of the input while increasing the depth of the feature representation. This downsampling process is crucial to capture the essential features necessary for discrimination.

Following the initial convolutional layer, the discriminator includes several additional convolutional layers, each designed to abstract further and refine the feature maps. These layers are interspersed with batch normalization to stabilize and accelerate the training process, and LeakyReLU activation functions to introduce non-linearity. The use of LeakyReLU helps mitigate the vanishing gradient problem, thereby facilitating the training of deeper networks and ensuring that the discriminator can learn complex patterns.

The final layer of the discriminator is a convolutional layer that maps the high-dimensional feature representation to a single scalar value. This scalar output is then passed through a Sigmoid activation function, which squashes the output to a range between 0 and 1, representing the probability of the input image being real. This probability score is used to compute the binary cross-entropy loss, which measures the discriminator's performance in distinguishing between real and generated images.

Overall, the discriminator architecture is designed to effectively evaluate the realism of the generated images by the generator. Through its convolutional layers, batch normalization, and activation functions, the discriminator learns to capture and differentiate the intricate features of real and generated MRI images. This adversarial interplay between the generator and discriminator ultimately drives the improvement of the generator's output quality, resulting in highly realistic and accurate medical image synthesis.

2) Training Process: The training process of the cGAN is a sophisticated and iterative procedure designed to train both the generator and discriminator networks simultaneously. The generator network, structured as a U-Net architecture, is tasked with producing realistic MRI images, while the

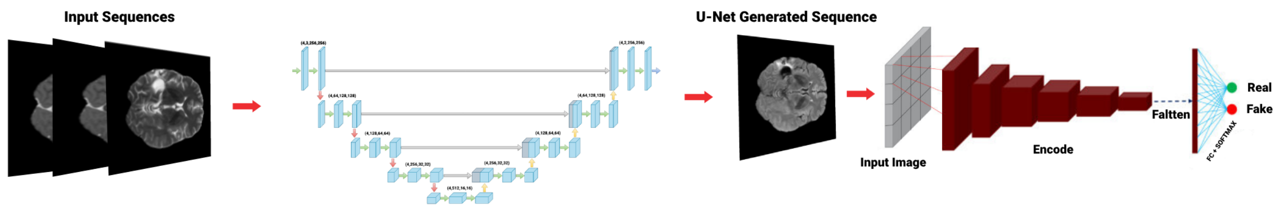

Fig. 1. Overall architecture of the proposed network (generator and discriminator).

discriminator network aims to distinguish between real and generated images. This adversarial training process is crucial for refining the networks' abilities.

Central to the training process is the use of a custom dataset class, which organizes the MRI images into input and target tensors. These tensors are efficiently loaded in batches using a DataLoader, ensuring a smooth and consistent data flow throughout the training process. The input modalities include T1n, T1c, and T2w images, while the target modalities encompass T2f images and segmentation masks.

The core of the training process involves simultaneously optimizing the generator and discriminator. The generator aims to minimize two key losses: the adversarial loss, which measures its ability to fool the discriminator, and the L1 loss, which quantifies the difference between the generated images and the real images. The discriminator, on the other hand, aims to maximize its ability to classify real and generated images accurately. This interplay between the generator and discriminator is what drives the improvement of the generator's output quality over time.

During each training iteration, the generator produces fake outputs, which the discriminator then evaluates. The discriminator's performance is measured using binary cross-entropy loss, which compares the discriminator's predictions against real and fake labels. The generator's performance is evaluated using a combination of adversarial loss and L1 loss, ensuring that the generated images are not only realistic but also closely match the real images.

The training process is monitored using metrics such as the Structural Similarity Index (SSIM) and Peak Signal-to-Noise Ratio (PSNR). These metrics provide quantitative measures of the similarity between the generated images and the real images, ensuring that the training process yields high-quality results. Throughout the training, the model's weights are periodically saved to allow for checkpointing and subsequent evaluation. This ensures that the best-performing models can be retained and further analyzed for their potential. The entire training process is a fine balance of generating realistic images and discerning their authenticity, ultimately leading to a robust and effective cGAN model.

3) Visualization and Analysis: A qualitative assessment of the generator's performance is conducted through the visualization of sample images. These visualizations include the input modalities, the ground truth images, and the generated images. The visual comparisons highlight the generator's ability to capture the essential features and structures present in the ground truth images, as demonstrated in Figure III-C3.

In conclusion, this methodology demonstrates the potential of cGANs for generating high-quality medical images. The proposed approach, which combines a U-Net generator and a PatchGAN discriminator, achieves promising results in both quantitative metrics and qualitative visualizations.

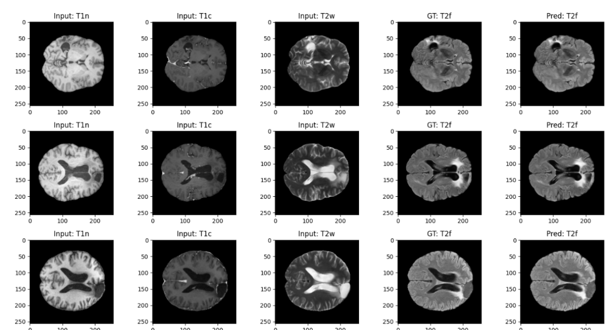

Fig. 2. Visual comparison of input MRI modalities and generated T2F images. From left to right: Input T1n, Input T1c, Input T2w, Ground Truth T2f, and Predicted T2f images. The generator network, a U-Net architecture, is trained to predict T2F images from the input modalities.

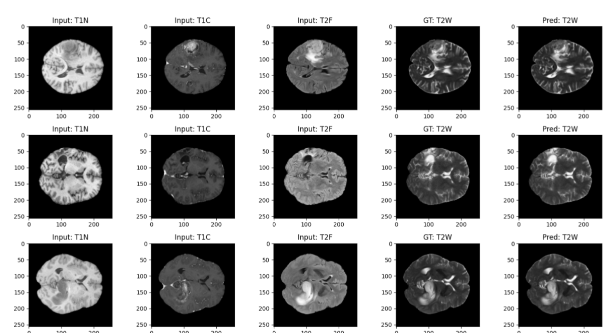

Fig. 3. Visual comparison of input MRI modalities and generated T2W images. From left to right: Input T1N, Input T1C, Input T2F, Ground Truth T2W, and Predicted T2W images. The generator network, a U-Net architecture, is trained to predict T2W images from the input modalities.

4) Evaluation Metrics: To evaluate the performance of the generator, we employ two widely used metrics: Structural Similarity Index (SSIM) and Peak Signal-to-Noise Ratio (PSNR).

SSIM measures the perceptual quality of the generated images by comparing their structural information with the real images. PSNR quantifies the ratio between the maximum possible power of a signal and the power of corrupting noise, providing an objective measure of image quality. These metrics are computed on a held-out test set, and the results demonstrate the effectiveness of the proposed approach in generating high-quality images. The SSIM and PSNR values indicate a high degree of similarity and low noise levels in the generated images, respectively.

TABLE I
COMPARISON OF DIFFERENT METHODS FOR THE GENERATION OF T2W AND T2F IMAGES BASED ON PSNR AND SSIM METRICS ON THE BRATS 2024 DATASET [10].

Method	**T2W**		**T2F**	
	PSNR	**SSIM**	**PSNR**	**SSIM**
pix2pix	26.155	0.821	×	×
MedGAN	25.616	0.805	×	×
pGAN	27.396	0.846	×	×
Transformer-Based Methods	27.535	0.861	×	0.859
ModaNet	**37.3918**	**0.9781**	×	**0.9692**

In the evaluation of image synthesis methods, the Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM) are crucial metrics for assessing the quality of generated images. As shown in Table I, ModaNet significantly outperforms the other methods in T2-weighted (T2W) image generation, achieving the highest PSNR of 37.3918 and SSIM of 0.9781. For T2-FLAIR (T2F) images, we specifically compare ModaNet with transformer-based methods using the SSIM metric. ModaNet achieves an SSIM of 0.9692, compared to 0.859 for the Transformer-based methods. This indicates that ModaNet produces images of superior quality and with greater similarity to the original images. The comparison underscores the superior performance of ModaNet in terms of image quality and structural fidelity, establishing it as a more effective method for tasks requiring high similarity to reference images.

IV. CONCLUSIONS

This paper presented ModaNet, a Conditional Generative Adversarial Network approach for synthesizing missing MRI modalities in brain tumor imaging. The methodology combines a U-Net generator with a patch-based discriminator to address the challenge of incomplete MRI sequences in clinical practice. Experimental results demonstrate superior performance compared to existing methods. ModaNet achieved a PSNR of 37.3918 and SSIM of 0.9781 for T2-weighted image generation, significantly outperforming pix2pix, MedGAN, and pGAN approaches. For T2-FLAIR synthesis, ModaNet attained an SSIM of 0.9692, exceeding transformer-based methods (0.859 SSIM). The evaluation on the BraTS 2024 dataset [10] validates the effectiveness of our approach in generating high-quality, realistic MRI sequences. The ability to synthesize missing modalities enhances diagnostic capabilities and supports AI-driven analysis in clinical settings where complete MRI sequences are unavailable. Future work includes evaluating generalizability across diverse datasets and conducting clinical validation studies. ModaNet represents a significant advancement in medical image synthesis, offering a valuable tool for improving brain tumor diagnosis and treatment planning.

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